

Question 1.

a. The Euler Equation

The social planner maximizes expected utility:

$$E_t \sum_{s=0}^{\infty} \beta^s \left[\frac{Z_{t+s}^c C_{t+s}^{1-\sigma}}{1-\sigma} \right]$$

subject to

$$C_t + I_t = A_t K_t^\alpha, \quad \text{With the production function : } Y_t = A_t K_t^\alpha$$

$$K_{t+1} = (1 - \delta)K_t + Z_t^x I_t. \quad (1)$$

This equation describes the law of motion of capital which shows that next period's capital stock is composed of undepreciated current capital and new investment which becomes more efficient in presence of IST.

From 1, we Solve for I_t :

$$I_t = \frac{K_{t+1} - (1 - \delta)K_t}{Z_t^x}. \quad (2)$$

In this model, investment is a flow variable that measures how much the next-period capital exceeds the undepreciated part of current capital, after adjusting for the IST shock. If $Z_t^x > 1$, the effective cost of current investment is reduced : less I_t is required to achieve a given level of future capital K_{t+1} . This relaxes the resource constraint and in equilibrium, households re-optimize their intertemporal allocation.

Substitute (2) into the resource constraint and solve for C_t :

$$C_t = A_t K_t^\alpha - \frac{K_{t+1} - (1 - \delta)K_t}{Z_t^x}.$$

A rise in Z_t^x lowers the relative price of investment and thus the consumption cost of capital : one unit of capital costs only $\frac{1}{Z_t^x}$ units of consumption. With investment now cheaper, households increase capital accumulation.

Let's solve the lagrangian :

$$\mathcal{L}_t = E_t \sum_{s=0}^{\infty} \beta^s \left[\frac{Z_{t+s}^c C_{t+s}^{1-\sigma}}{1-\sigma} + \lambda_{t+s} \left(A_{t+s} K_{t+s}^\alpha - C_{t+s} - \frac{K_{t+s+1} - (1 - \delta)K_{t+s}}{Z_{t+s}^x} \right) \right].$$

First Order Condition with respect to. C_t :

$$\frac{\partial \mathcal{L}_t}{\partial C_t} = Z_t^c C_t^{-\sigma} - \lambda_t = 0 \quad \Rightarrow \quad \lambda_t = Z_t^c C_t^{-\sigma}. \quad (3)$$

λ_t is the current value (in utility terms) of one unit of resources available in period t .

First Order Condition with respect to K_{t+1} :

$$\begin{aligned}\frac{\partial \mathcal{L}_t}{\partial K_{t+1}} &= -\beta^t \frac{\lambda_t}{Z_t^x} + \beta^{t+1} E_t \left[\lambda_{t+1} \left(A_{t+1} \alpha K_{t+1}^{\alpha-1} + \frac{1-\delta}{Z_{t+1}^x} \right) \right] = 0. \\ \Rightarrow \frac{\lambda_t}{Z_t^x} &= \beta E_t \left[\lambda_{t+1} \left(A_{t+1} \alpha K_{t+1}^{\alpha-1} + \frac{1-\delta}{Z_{t+1}^x} \right) \right].\end{aligned}$$

This optimality condition states that the household chooses K_{t+1} so that the marginal utility cost of investing today (λ_t/Z_t^x) is exactly equal to the expected discounted marginal utility benefit, that is, the utility value of the undepreciated capital plus its net return..

Substituting $\lambda_t = Z_t^c C_t^{-\sigma}$ and $\lambda_{t+1} = Z_{t+1}^c C_{t+1}^{-\sigma}$ yields the Euler equation:

$$Z_t^c C_t^{-\sigma} \frac{1}{Z_t^x} = \beta E_t \left[Z_{t+1}^c C_{t+1}^{-\sigma} \left(A_{t+1} \alpha K_{t+1}^{\alpha-1} + \frac{1-\delta}{Z_{t+1}^x} \right) \right].$$

Economic Intuition: This Euler equation shows how households choose between consuming today and saving to consume more in the future. Giving up one unit of consumption today means investing it in capital, which tomorrow yields the expected discounted marginal product of capital plus the undepreciated part of the investment, adjusted for investment efficiency. Households choose their consumption path so that the marginal utility cost of forgoing consumption today equals the discounted expected marginal utility of the gross return tomorrow, making them indifferent between both options. When the return on saving is greater than how much the household discounts the future, the household chooses to raise consumption over time.

In this model, preference and IST shocks contribute to the trade-off. When the expected future preference shock Z_{t+1}^c is large to the current one Z_t^c , households put more weight on future utility and are more willing to cut current consumption and invest. When Z_t^c is relatively high, current utility is weighted more, so they consume more now and save less. A current higher IST shock lowers the marginal cost of creating capital today (each unit of forgone consumption produces more effective capital), which strengthens the incentive to reduce current consumption and invest more.

b) Derivation of Steady-State Variables

In a (non-stochastic) steady state, we have:

$$C_t = \bar{C}, \quad I_t = \bar{I}, \quad K_t = \bar{K}, \quad A_t = \bar{A}, \quad Z_t^c = \bar{Z}^c, \quad Z_t^x = \bar{Z}^x.$$

(i) Steady-state capital accumulation.

$$K_{t+1} = (1 - \delta)K_t + Z_t^x I_t$$

Since at steady state variables stop changing over time and become constant regardless of the period, we have :

$$\begin{aligned} K_{t+1} &= K_t = \bar{K} \\ \bar{K} &= (1 - \delta)\bar{K} + \bar{Z}^x \bar{I} \\ \bar{I} &= \frac{\delta}{\bar{Z}^x} \bar{K}. \end{aligned} \tag{4}$$

In steady state, since capital is constant, investment only needs to replace the depreciated capital adjusted for IST. when IST is higher, each unit of investment is more efficient, so less investment is required to offset depreciation. Replacing the depreciated part, adjusted for IST, is enough to keep the capital stock unchanged across periods.

(ii) Steady-state resource constraint.

$$C_t + I_t = A_t K_t^\alpha$$

In steady state :

$$\bar{C} + \bar{I} = \bar{A} \bar{K}^\alpha$$

We substitute $\bar{I} = \frac{\delta}{\bar{Z}^x} \bar{K}$ and solve for $\bar{C}$

$$\bar{C} = \bar{A} \bar{K}^\alpha - \frac{\delta}{\bar{Z}^x} \bar{K}.$$

Regarding the previous comment, in steady state, consumption equals output minus the depreciated part of capital, adjusted for IST.

(iii) Steady-state Euler equation. We have :

$$\frac{Z_t^c}{Z_t^x} C_t^{-\sigma} = \beta E_t \left[Z_{t+1}^c C_{t+1}^{-\sigma} \left(A_{t+1} \alpha K_{t+1}^{\alpha-1} + \frac{1 - \delta}{Z_{t+1}^x} \right) \right].$$

In steady state, variables no longer vary over time, so the time subscripts can be dropped.:

$$\frac{\bar{Z}^c}{\bar{Z}^x} \bar{C}^{-\sigma} = \beta \bar{Z}^c \bar{C}^{-\sigma} \left(\bar{A} \alpha \bar{K}^{\alpha-1} + \frac{1 - \delta}{\bar{Z}^x} \right).$$

cancel $\bar{Z}^c \bar{C}^{-\sigma} \neq 0$:

$$\frac{1}{\bar{Z}^x} = \beta \left(\bar{A}\alpha \bar{K}^{\alpha-1} + \frac{1-\delta}{\bar{Z}^x} \right).$$

In steady state, the euler equation shows that the marginal cost of creating one unit of effective capital, given by $1/\bar{Z}^x$, must equal the discounted marginal benefit of that capital, which comes from the marginal product of capital and the undepreciated part of the capital stock. A higher IST reduces the cost of generating new effective capital, but it also increases the obsolescence of existing capital because new investment comes with better technology and becomes relatively more productive than old capital.

Rearrange and then solve for $\bar{K}$:

$$1 = \beta \left(\bar{Z}^x \bar{A}\alpha \bar{K}^{\alpha-1} + 1 - \delta \right).$$

$$\beta \bar{Z}^x \bar{A}\alpha \bar{K}^{\alpha-1} = 1 - \beta(1 - \delta),$$

$$\bar{K}^{\alpha-1} = \frac{1 - \beta(1 - \delta)}{\beta \bar{Z}^x \bar{A}\alpha},$$

$$\bar{K} = \left[\frac{1 - \beta(1 - \delta)}{\beta \bar{Z}^x \bar{A}\alpha} \right]^{\frac{1}{\alpha-1}}.$$

Though parameters are supposed to be fixed, they can vary by country, let us check the expected signs on $\bar{K}$ after some algebra :

$$\frac{\partial \bar{K}}{\partial \beta} > 0, \quad \frac{\partial \bar{K}}{\partial \delta} < 0, \quad \frac{\partial \bar{K}}{\partial \alpha} > 0.$$

More patient countries value future consumption more, so they invest more and reach a higher steady-state capital. A higher depreciation rate yields in low steady-state capital, while a production function with a higher capital share leads to a higher steady-state level of capital.

Substituting the expression for $\bar{K}$, we obtain the steady-state investment

$$\bar{I} = \frac{\delta}{\bar{Z}^x} \left[\frac{1 - \beta(1 - \delta)}{\beta \bar{Z}^x \bar{A}\alpha} \right]^{\frac{1}{\alpha-1}}.$$

After some calculations, in this model countries with higher β and production structures with higher α will exhibit higher steady state investment. Only the sign of δ is ambiguous : in equation 4, a higher depreciation rate directly increases the investment needed to replace depreciated capital, but indirectly reduces investment by lowering the steady-state capital stock.

The same, the steady-state consumption

$$\bar{C} = \bar{A} \left[\frac{1 - \beta(1 - \delta)}{\beta \bar{Z}^x \bar{A} \alpha} \right]^{\frac{\alpha}{\alpha-1}} - \frac{\delta}{\bar{Z}^x} \left[\frac{1 - \beta(1 - \delta)}{\beta \bar{Z}^x \bar{A} \alpha} \right]^{\frac{1}{\alpha-1}}.$$

After algebra, we find countries with higher β may have higher steady-state consumption : greater patience leads to more saving, a larger capital stock, expands production, and ultimately allows more resources for both investment and consumption. Therefore, more patient countries may enjoy structural economic advantages.

c) Codes dynare

The representative firm will maximize the profit.

$$\max \pi_t = A_t k_t^\alpha - w_t - r_t k_t - \delta_t k_t,$$

Rental Rate of Capital : The derivative gives

$$\frac{\partial \pi_t}{\partial r_t} = \alpha A_t k_t^{\alpha-1} - \delta_t - r_t = 0$$

$$r_t = \alpha A_t k_t^{\alpha-1} - \delta_t$$

$$r_t = MPK_t - \delta_t$$

This the net rental rate of capital.

Replacement price of capital :

$$\delta_t = 1 - \frac{1 - \delta}{Z_t^x} \quad (5)$$

In presence of IST, there is a replacement price of capital which is the obsolescence (if $Z_t^x > 1$) of existing capital as new investment becomes relatively more productive.

The gross return on capital is $(1 + \bar{r}_t)$ which is the net return and the invested capital :

$$1 + \bar{r}_t = r_t + \frac{1 - \delta}{Z_t^x}$$

Implicit wage :

$$y_t = A_t k_t^\alpha l_t^{1-\alpha}$$

$$\max \pi_t = A_t k_t^\alpha l_t^{1-\alpha} - l_t w_t - r_t k_t - \delta_t k_t$$

$$\frac{\partial \pi_t}{\partial l_t} = A_t k_t^\alpha (1 - \alpha) l_t^{-\alpha} - w_t = 0$$

Since l is inelastic, we can replace it by 1 then we have :

$$w_t = (1 - \alpha) y_t \quad (6)$$

We have considered variables including the replacement rate of capital (repla) to evaluate the effect of the IST shock on both the replacement rate of capital and the implicit rental price. In the code, we worked with the net rental price rather than the gross rental price, since their patterns are almost the same.

Codes

```
// Summarative assignment

var c k y inv w r repla a zx zc;
//With R the replacement price of capital
// 3 exogenous shocks: tfp, investment-specific technology, preference
varexo eps_a eps_zx eps_zc;

parameters

beta alpha sigma delta rho_a rho_zx rho_zc abar zxbar zcbar;

abbar = 1;
zxbar = 1;
zcbar = 1;
beta = 0.99;
delta = 0.02;
rho_a = 0.9;
rho_zx = 0.9;
rho_zc = 0.9;
alpha = 0.35;
sigma = 1.5;

// Exact equation system
model;

// Euler equation
zc*c^(-sigma)*(1/zx)
= beta*(zc(+1)*c(+1)^(-sigma)*
(a(+1)*alpha*k^(alpha-1) + (1-delta)/zx(+1)));
```

```

// Resource constraint
c + (k - (1-delta)*k(-1))/zx = a*k(-1)^alpha;

// GDP
y = a*k(-1)^alpha;

// Investment
inv = (k - (1-delta)*k(-1))/zx;

// Wage
w = (1-alpha)*y;

//replacement price of capital
repla = 1- ((1-delta)/zx);

// Rental price (net)
r = alpha*a*k(-1)^(alpha-1) - repla;

// Stochastic processes
zc - zcbar = rho_zc*(zc(-1) - zcbar) + eps_zc;
zx - zxbar = rho_zx*(zx(-1) - zxbar) + eps_zx;
a - abar = rho_a*(a(-1) - abar) + eps_a;

end;

steady_state_model;

a = abar;
zc = zcbar;
zx = zxbar;

k = ((1 - beta*(1-delta)) / (alpha*beta*a*zx))^(1/(alpha-1));
c = a*k^alpha - (delta/zx)*k;
y = a*k^alpha;
inv = (delta/zx)*k;
w = (1-alpha)*y;
repla = 1- ((1-delta)/zx);
r = alpha*a*k^(alpha-1) - repla;

```

```

end;

steady;
check;

shocks;
var eps_a; stderr 0.01;
var eps_zx; stderr 0.01;
var eps_zc; stderr 0.01;
end;

stoch_simul(irf = 100, order = 1) c k y inv w r repla a zx zc ;

```

Output

```

>> dynare S11
Starting Dynare (version 6.2).
Calling Dynare with arguments: none
Starting preprocessing of the model file ...
Found 10 equation(s).
Evaluating expressions...
Computing static model derivatives (order 1).
Normalizing the static model...
Finding the optimal block decomposition of the static model...
6 block(s) found:
5 recursive block(s) and 1 simultaneous block(s).
the largest simultaneous block has 2 equation(s)
and 2 feedback variable(s).
Computing dynamic model derivatives (order 1).
Normalizing the dynamic model...
Finding the optimal block decomposition of the dynamic model...
6 block(s) found:
5 recursive block(s) and 1 simultaneous block(s).
the largest simultaneous block has 2 equation(s)
and 2 feedback variable(s).

```

Preprocessing completed.
Preprocessing time: 0h00m01s.

Steady-state results

Table 1: steady-state values

variable	steady state
c	2.8759
k	43.5723
y	3.74735
inv	0.871446
w	2.43577
r	0.010101
repla	0.0200
a	1.0000
zx	1.0000
zc	1.0000

Eigenvalues

Table 2: eigenvalues

	modulus	real	imaginary
1	0.9	0.9	0.0
2	0.9	0.9	0.0
3	0.9	0.9	0.0
4	0.9758	0.9758	0.0
5	1.035	1.035	0.0
6	6.944e+16	6.944e+16	0.0
7	6.944e+16	6.944e+16	0.0
8	6.944e+16	6.944e+16	0.0

There are 4 eigenvalue(s) larger than 1 in modulus for 4 forward-looking variable(s)
The rank condition is verified.

Model summary

Table 3: model summary

item	value
number of variables	10
number of stochastic shocks	3
number of state variables	4
number of jumpers	4
number of static variables	5

Matrix of covariance of exogenous shocks

Table 4: covariance matrix of exogenous shocks

variables	eps_a	eps_zx	eps_zc
eps_a	0.000100	0.000000	0.000000
eps_zx	0.000000	0.000100	0.000000
eps_zc	0.000000	0.000000	0.000100

Policy and transition functions

Table 5: policy and transition functions

	c	k	y	inv	w	r	repla	a	zx	zc
constant	2.875899	43.572325	3.747346	0.871446	2.435775	0.010101	0.020000	1.000000	1.000000	1.000000
$k(-1)$	0.034328	0.975773	0.030101	-0.004227	0.019566	-0.000449	0.000000	0.000000	0.000000	0.000000
$a(-1)$	0.535365	2.837247	3.372611	2.837247	2.192197	0.027091	0.000000	0.900000	0.000000	0.000000
$zx(-1)$	-1.414709	2.199011	0.000000	1.414709	0.000000	-0.882000	0.882000	0.000000	0.900000	0.000000
$zc(-1)$	1.276472	-1.276472	0.000000	-1.276472	0.000000	0.000000	0.000000	0.000000	0.000000	0.900000
eps_a	0.594850	3.152496	3.747346	3.152496	2.435775	0.030101	0.000000	1.000000	0.000000	0.000000
eps_zx	-1.571899	2.443345	0.000000	1.571899	0.000000	-0.980000	0.980000	0.000000	1.000000	0.000000
eps_zc	1.418302	-1.418302	0.000000	-1.418302	0.000000	0.000000	0.000000	0.000000	0.000000	1.000000

Theoretical moments (volatility)

Table 6: theoretical moments

variable	mean	std. dev.	variance	cv (%) (added)
c	2.8759	0.0712	0.0051	2.48
k	43.5723	1.7430	3.0382	4.00
y	3.7473	0.1136	0.0129	3.03
inv	0.8714	0.0844	0.0071	9.69
w	2.4358	0.0739	0.0055	3.03
r	0.0101	0.0227	0.0005	8.91
repla	0.0200	0.0225	0.0005	-
a	1.0000	0.0229	0.0005	-
zx	1.0000	0.0229	0.0005	-
zc	1.0000	0.0229	0.0005	-

Variance decomposition

Table 7: variance decomposition (in percent)

variable	eps_a	eps_zx	eps_zc
c	52.78	28.99	18.24
k	55.46	33.31	11.23
y	90.50	7.10	2.39
inv	69.25	16.73	14.02
w	90.50	7.10	2.39
r	0.09	99.89	0.01
repla	0.00	100.00	0.00
a	100.00	0.00	0.00
zx	0.00	100.00	0.00
zc	0.00	0.00	100.00

Correlation matrix

Table 8: matrix of correlations

variables	c	k	y	inv	w	r	repla	a	zx	zc
c	1.0000	0.7012	0.6709	0.0600	0.6709	0.2932	-0.3070	0.4496	-0.3070	0.3413
k	0.7012	1.0000	0.7189	0.3766	0.7189	-0.2858	0.2640	0.3406	0.2640	-0.1533
y	0.6709	0.7189	1.0000	0.7805	1.0000	-0.1053	0.1097	0.8982	0.1097	-0.0637
inv	0.0600	0.3766	0.7805	1.0000	0.7805	-0.3890	0.4065	0.8301	0.4065	-0.3734
w	0.6709	0.7189	1.0000	0.7805	1.0000	-0.1053	0.1097	0.8982	0.1097	-0.0637
r	0.2932	-0.2858	-0.1053	-0.3890	-0.1053	1.0000	-0.9993	0.0199	-0.9993	0.0048
repla	-0.3070	0.2640	0.1097	0.4065	0.1097	-0.9993	1.0000	0.0000	1.0000	0.0000
a	0.4496	0.3406	0.8982	0.8301	0.8982	0.0199	0.0000	1.0000	0.0000	0.0000
zx	-0.3070	0.2640	0.1097	0.4065	0.1097	-0.9993	1.0000	0.0000	1.0000	0.0000
zc	0.3413	-0.1533	-0.0637	-0.3734	-0.0637	0.0048	0.0000	0.0000	0.0000	1.0000

Autocorrelation coefficients

Table 9: Coefficients of autocorrelation

Order	1	2	3	4	5
c	0.9477	0.8994	0.8548	0.8136	0.7755
k	0.9987	0.9952	0.9896	0.9824	0.9736
y	0.9437	0.8919	0.8443	0.8004	0.7600
inv	0.8931	0.7970	0.7107	0.6332	0.5636
w	0.9437	0.8919	0.8443	0.8004	0.7600
r	0.9018	0.8133	0.7337	0.6619	0.5973
repla	0.9000	0.8100	0.7290	0.6561	0.5905
a	0.9000	0.8100	0.7290	0.6561	0.5905
zx	0.9000	0.8100	0.7290	0.6561	0.5905
zc	0.9000	0.8100	0.7290	0.6561	0.5905

Total computing time : 0h00m12s

Figure 1: orthogonalized impulse responses to a productivity shock

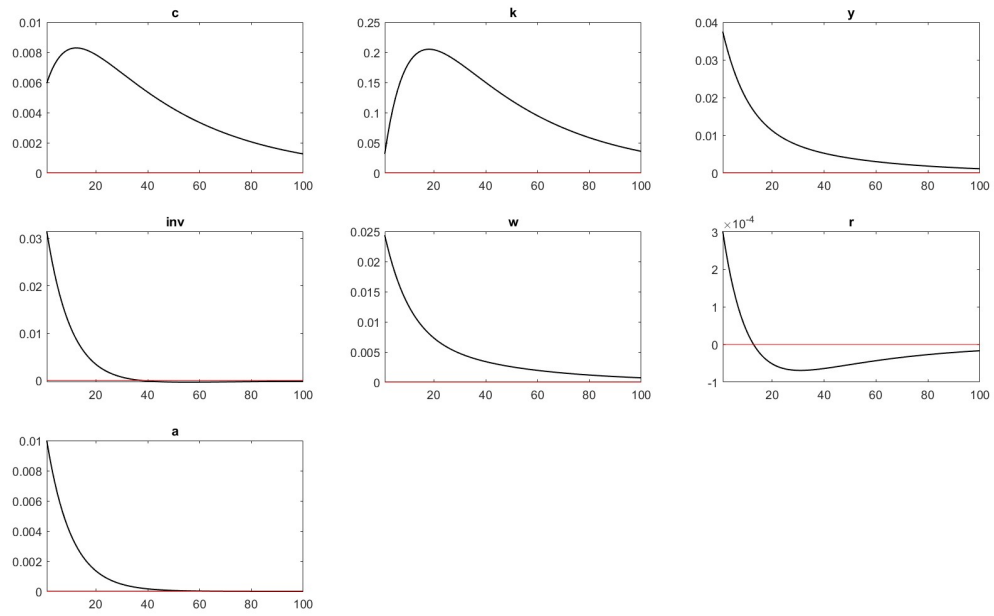


Figure 2: orthogonalized impulse responses to investment specific technology shock

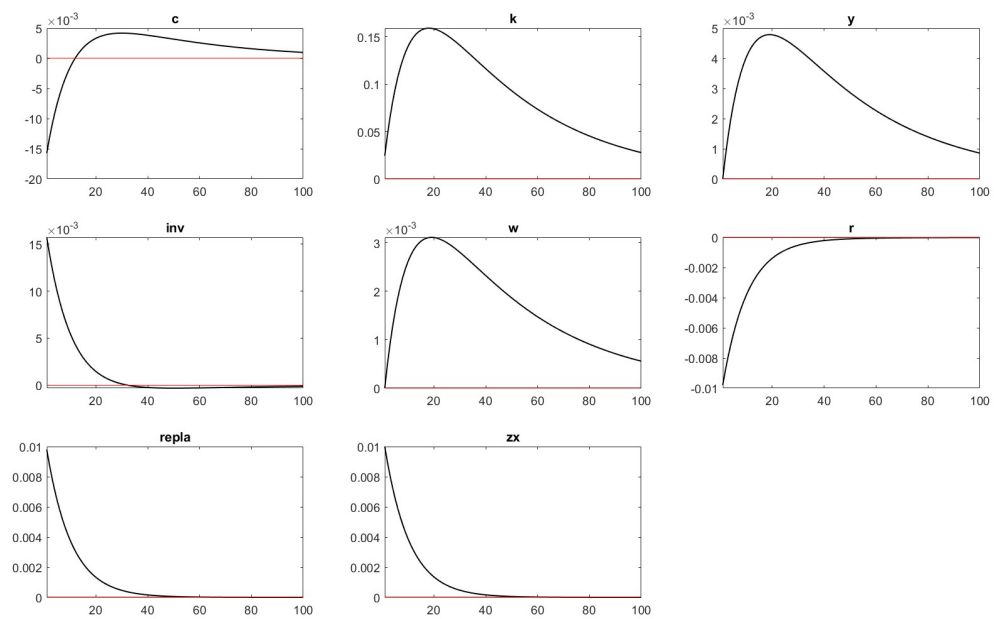
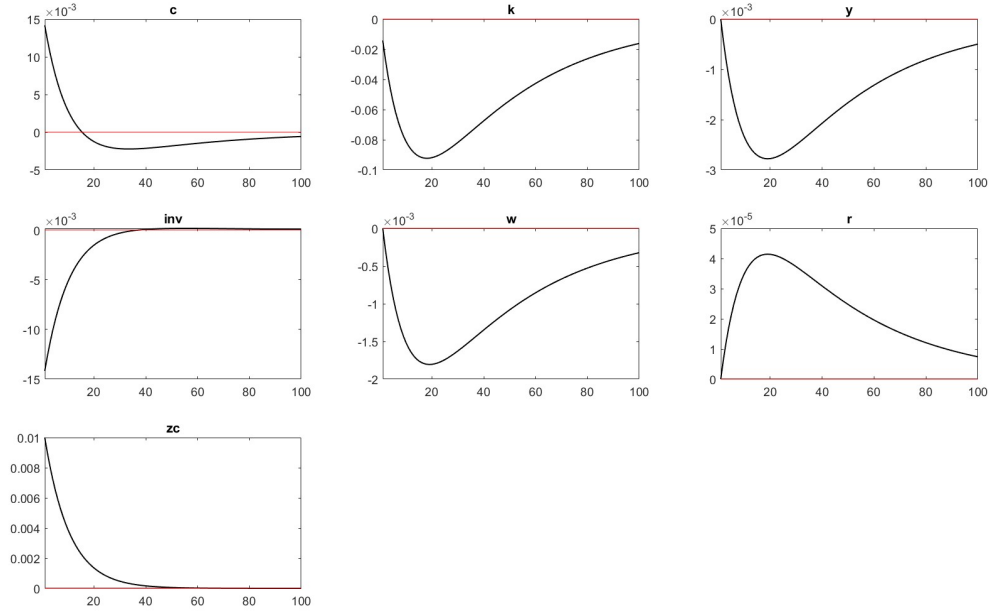


Figure 3: orthogonalized impulse responses to preference shock



d) Project report

The blanchard–kahn condition holds because the number of eigenvalues above one (table 2) equals the number of forward-looking variables, ensuring a unique equilibrium and convergence to the steady state.

Steady-State (table 1)

The capital has the highest steady-state (43.6) because it is a stock and δ is low (0.02). At steady state, with $Z_t^x = 1$, equation (5) implies $\delta_t = \delta$. Steady-state consumption is 2.876 and output is 3.75, so consumption is 77% of output, the rest being investment.

Theoretical Moments (Volatility) (table 6)

As in [Hodrick et al. \(1991\)](#), for comparison we use relative volatility (σ/μ) computed in the last column (table 6), as it is insensitive to steady-state magnitudes. Investment is the most volatile variable (9.7%). Output fluctuates at about one-third of that (3.03%), and consumption remains smooth (2.5%). These results align with RBC theory and the classical statement in [Kydland and Prescott \(1982\)](#): “*Investment varies three times as much as output.*” They also match the volatility patterns reported in [Plosser \(1989\)](#) and reproduced in the lecture notes ([Basu, 2025](#)). Consumption is smooth because households weigh the utility cost of reducing current consumption against expected investment

returns.

Variance decomposition (table 7)

Productivity shock. Our results confirm the classic RBC result that the TFP shock is the main driver of business cycles. It explains over 90% of output and wage volatility, more than half of consumption and capital volatility, and 69.25% of investment. This dominance comes from the multiplicative role of productivity in production and wage equations. With $\alpha = 0.35$, the term $(1-\alpha)$ is still large, which strengthens wage responses.

IST shock and Preference shock. The IST shock explains 28.99% of consumption volatility and about one third of capital fluctuations. It enters both the euler and investment equations, shaping intertemporal choices and capital accumulation. In contrast, IST and preference shocks have smaller effects on output (7.1% and 2.4%) because they mainly influence intertemporal decisions.

The rental price of capital is almost fully driven by the IST shock (99.89%) because it moves nearly one-to-one with the replacement price of capital, directly linked to Z_t^x . A higher Z_t^x accelerates capital obsolescence, lowering the value of existing capital and thus reducing the net return.

The preference shock has overall smaller impact because prices are flexible and households are risk-averse, so they smooth consumption as they don't like higher intertemporal fluctuations. This is why it accounts for only about 18% of consumption volatility.

Correlation (table 8)

Our results match business-cycle stylized facts: during expansions, output increases with consumption (0.67), investment (0.78), capital (0.72), and wages (1). These correlations align with Plosser (1989), Panel B “Predicted”, supporting the RBC view that most macro variables are strongly procyclical.

The perfect wage–output correlation arises because the implicit wage is a linear function of the MPL under flexible prices and no frictions. This is unrealistic, as real wages and prices are not fully flexible; even in Plosser (1989), Panel A “Actual”, the observed correlation is below 0.60.

Consumption is less correlated with output than investment because of smoothing. The diminishing MPK explains the negative correlation between r and K (-0.286).

Autocorrelation

The autocorrelation pattern shows that, overall all variables are highly persistent, consistent with standard RBC results ?. Capital is the most persistent variable, with an

autocorrelation of 0.99 at lag one and 0.97 at lag five, reflecting its AR(1) structure. We have Simulated with a higher δ which reduced persistence only slightly, confirming that the AR component dominates. The high discount factor ($\beta = 0.99$) makes households very patient, yielding smoother and more persistent consumption. Output and wages are also persistent.

Impulse Response Function and Policy and Transition Function

In figure 1, Following a positive TFP shock, the marginal products of labor and capital rise, raising output, wages, and the rental price. Households become richer and allocate the extra income between consumption and investment through the euler equation and the resource constraint. Investment, the most volatile and least persistent variable, reacts sharply and returns quickly because it adjusts the resource constraint when consumption is smoothed. Output and wages move monotonically because they depend directly on the TFP shock. Consumption adjusts smoothly with a hump-shaped pattern due to intertemporal optimization. Capital rise is due to investment increase and follows a smooth path because its AR process. The GDP increases by about 0.0375, roughly 1% of its steady state level. Similar calculations for all variables show that response magnitudes reflect their relative volatilities (table 6), their relationship to the shock, and the model's structure.

In figure 2, A positive IST shock raises the efficiency of new investment, reducing the marginal cost of accumulating capital. With cheaper investment, households shift resources from consumption to investment. Consumption falls, investment rises immediately influencing capital increase gradually. More capital leads to higher production.

Because new investment becomes more efficient, existing capital loses relative value, increasing the replacement price and lowering the implicit rental rate. The larger capital stock also faces diminishing returns, further reducing the rental rate.

With flexible prices, adjustments occur mainly through prices. More efficient capital raises workers' productivity, increasing the marginal value of labor and encouraging firms to offer higher wages. Higher production also raises the MPL, reinforcing wage increases.

In figure 3, A positive preference shock makes households value current consumption more, reducing saving and investment. Lower investment slows capital accumulation, leading the capital stock to decline. Then, the MPK rises, increasing the rental price. With less capital, production falls and wages decline. However, the policy transition matrix shows that preference and IST shocks have limited effects on output and wages because they do not directly affect marginal products.

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